Can Prediction-Powered Confidence Intervals for a Mean Be Shortened with Everything Held Fixed? A Negative Result on Real Data and a Characterisation

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Abstract

With a small labelled set, a large unlabelled set and a frozen predictor, prediction-powered inference (PPI) and PPI++ give valid intervals that are much shorter than classical ones. We ask whether, with none of these three inputs changed, the interval for a population mean can be shortened further. The route we test is to recalibrate the frozen predictor with a cross-fitted regression of the label on the prediction (and, where available, covariates), fitted on the labelled data. We derive the asymptotic variance of the resulting estimator: the gain over PPI++ is governed by the gap between the correlation ratio η^2 of Y on the prediction and the squared linear correlation ρ^2. On six real datasets (3 sample sizes, 1000 replications each) recalibration did *not* shorten the interval: widths were equal to PPI++ at $n = 1000$ and larger at small n (up to 14% for isotonic, up to $2\times$ for boosted trees). On synthetic data with a non-monotone calibration curve, the same method halves the width (0.065 vs 0.131 at $n = 2000$), as theory predicts. PPI++ itself under-covers at $n = 50$ (down to 81%), and a t critical value only partly repairs this. So with off-the-shelf predictors whose calibration curves are close to linear, PPI++ is essentially not improvable by recalibration; gains need strongly nonlinear miscalibration and enough labels to learn it.

1 Introduction

Scientists often have n gold-standard labels, $N \gg n$ unlabelled points, and a fixed pre-trained predictor f. Prediction-powered inference [Angelopoulos et al., 2023] and PPI++ [Angelopoulos et al., 2023] turn f into valid intervals, and cross-prediction [Zrnic and Candès, 2023] handles predictors trained on the labelled data. The question here is narrower: *with labels, unlabelled data and f all fixed, can the nominal-coverage interval still be shortened, by how much, and when?*

We study the mean $\theta = \mathbb{E}[Y]$. We chose it because the PPI++ estimator and its optimal tuning have closed form, so any improvement can be traced to a specific mechanism, and because it is the target of most examples in the PPI literature. The only freedom left is how to use the labelled data: (i) the tuning weight λ (already optimised in PPI++), (ii) the finite-sample critical value, and (iii) replacing f with a better function $g(f(X), X)$ learned on the labelled data. We test (ii) and (iii).

Contributions. (1) A variance characterisation (Prop. 1) showing exactly when route (iii) can help and by how much. (2) A large replicated benchmark on six real datasets against classical, PPI, PPI++, comparing widths and coverage. (3) A mostly negative finding: on these datasets recalibration gives no width gain, and PPI++ has a small-n coverage deficit. (4) Synthetic settings that map where the gain does and does not exist. We make no claim of a new asymptotic optimality result; the estimator

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is a cross-fitted PPI++ with a learned g, a known idea in the spirit of cross-prediction and debiased ML [Zrnic and Candès, 2023, Chernozhukov et al., 2016].

2 Related work

PPI [Angelopoulos et al., 2023] forms $\hat{\theta} = \bar{Y}_{\text{lab}} - \overline{(f-)}$-type rectified estimates $\frac{1}{N}\sum f(\tilde{X}_j) + \frac{1}{n}\sum(Y_i - f(X_i))$; PPI++ [Angelopoulos et al., 2023] scales f by a data-driven λ so it never does worse than classical asymptotically. Cross-prediction [Zrnic and Candès, 2023] cross-fits a predictor on the labelled data. Active inference [Zrnic and Candès, 2024] and stratified PPI [Fisch et al., 2024] change the labelling or use strata, i.e. change the inputs we hold fixed. Double machine learning [Chernozhukov et al., 2016] supplies the orthogonality idea behind using a learned nuisance with a debiasing term. Datasets come from the PPI package, with original sources including Willett et al. [2013].

3 Setting and method

Data: labelled $(X_i, Y_i)_{i\le n}$ i.i.d., unlabelled $\tilde{X}_j$, $j \le N$, same distribution, $r = n/N$. Write $F = f(X)$.

Baselines. Classical: $\bar{Y} \pm z\, s/\sqrt{n}$. PPI: $\lambda = 1$. PPI++: for a function g and λ,

$$\hat{\theta}_\lambda = \frac{1}{n}\sum_i (Y_i - \lambda g_i) + \frac{\lambda}{N}\sum_j g(\tilde{X}_j), \quad \hat{\sigma}^2 = \frac{\widehat{\mathrm{Var}}(Y-\lambda g)}{n} + \lambda^2\frac{\widehat{\mathrm{Var}}(g(\tilde{X}))}{N},$$

interval $\hat{\theta}_\lambda \pm z_{0.975}\hat{\sigma}$; $\hat{\lambda} = \widehat{\mathrm{Cov}}(Y, g)/((1+r)\widehat{\mathrm{Var}}(g))$ using the pooled variance of g. PPI++ uses $g = f$. We also report PPI++ with a t_{n-1} critical value (PPI++_t).

Recalibrated PPI++ (recal). Split the labelled set in $K = 5$ folds; fit $\hat{g}_{-k}$ on the other folds; use $g_i = \hat{g}_{-k(i)}(F_i)$ for labelled points and $g(\tilde{X}_j) = \frac{1}{K}\sum_k \hat{g}_{-k}(\tilde{X}_j)$ for unlabelled points, then apply the PPI++ formulas above. The $\hat{g}$ is either isotonic regression of Y on F (iso) or a small gradient-boosted tree regressor (gbm; 60 iterations, 6 leaves, min 20 per leaf) on F (plus the covariates in the census sets). The $\hat{\sigma}$ ignores the dependence across folds; coverage is therefore checked empirically.

Proposition 1 (asymptotic widths). Assume f (or g) is a fixed square-integrable function, $\mathrm{Var}(Y), \mathrm{Var}(g) \in (0, \infty)$, and $n/N \to r$. Then $\sqrt{n}(\hat{\theta}_\lambda - \theta) \Rightarrow N(0, V(\lambda))$ with $V(\lambda) = \mathrm{Var}(Y - \lambda g) + \lambda^2 r\,\mathrm{Var}(g)$, minimised at $\lambda^* = \mathrm{Cov}(Y, g)/((1+r)\,\mathrm{Var}\, g)$ with $V^* = \mathrm{Var}\, Y - \mathrm{Cov}(Y, g)^2/((1+r)\,\mathrm{Var}\, g)$. (a) For $g = f$: $V^* = \mathrm{Var}\, Y\,[1 - \rho^2/(1+r)]$, $\rho = \mathrm{Corr}(Y, f)$. (b) For $g = m(Z) = \mathbb{E}[Y \mid Z]$ with $Z = (F, \text{covariates})$: $\mathrm{Cov}(Y, g) = \mathrm{Var}\, g$, so $V^* = \mathrm{Var}\, Y\,[1 - \eta^2/(1+r)]$ with $\eta^2 = \mathrm{Var}\, m(Z)/\mathrm{Var}\, Y \ge \rho^2$. *Proof.* Independent sample means with the CLT; V is a quadratic in λ, and $\mathrm{Cov}(Y, m) = \mathbb{E}[m\,\mathbb{E}(Y|Z)] - \ldots = \mathrm{Var}\, m$ by the tower property. (If f is a function of Z then $\eta^2 \ge \rho^2$ by Cauchy–Schwarz.) □

So the best possible width ratio of (b) to (a) is $\sqrt{[1 - \eta^2/(1+r)]/[1 - \rho^2/(1+r)]}$, an *upper limit on gains*; the learned $\hat{g}$ pays an estimation price of order $1/n$ in variance (more for flexible learners), which is why gains need both $\eta^2 - \rho^2$ large and n large. The oracle variance is not attained with estimated $\hat{g}$ and we do not prove validity of the fold-ignoring $\hat{\sigma}$.

4 Experiments

Real data. Six sets from `ppi-python` 0.2.3 [Angelopoulos et al., 2023]: galaxies, forest, alphafold, gene_expression, census_income, census_healthcare (`Y`,`Yhat`; census also `X`). Plankton and ballots were not used. Target: mean of Y. The empirical distribution of the full dataset is treated as the population, so the truth is its mean; each replication draws $n \in \{50, 200, 1000\}$ labelled and $N = 5000$ unlabelled points *with replacement* from it (so draws are exactly i.i.d. from this population). 1000 replications per cell, seeds in the code. Level 95%. Binomial s.e. of a coverage estimate is about 0.7 points near 95%. Table 1 reports coverage (%) and average width relative to PPI++ (so PPI++ = 1).

Table 1: Real data: coverage (%) and mean width relative to PPI++ (same column, w), 1000 replications, nominal 95%. Script: `code/run.py`, `code/analyze.py`.

data	n	classical		PPI		PPI++		PPI++_t		recal-iso		recal-iso_t		recal-gbm		recal-gbm_t	
		cov	w	cov	w	cov	w	cov	w	cov	w	cov	w	cov	w	cov	w
galaxies	50	95.1	1.465	93.5	0.985	91.3	1.000	92.1	1.025	91.1	1.074	92.0	1.101	93.2	1.290	93.6	1.322
galaxies	200	95.1	1.456	94.9	0.999	94.4	1.000	94.5	1.006	94.6	1.029	94.6	1.035	94.1	1.047	94.4	1.054
galaxies	1000	94.8	1.358	95.4	1.021	95.8	1.000	95.9	1.001	95.5	1.006	95.5	1.008	95.3	1.014	95.3	1.015
forest	50	93.7	1.166	91.9	0.983	89.8	1.000	90.5	1.025	90.6	1.056	90.9	1.082	92.2	1.496	92.9	1.534
forest	200	95.3	1.175	93.7	1.000	93.5	1.000	93.6	1.006	93.4	1.016	93.4	1.022	94.3	1.061	94.7	1.068
forest	1000	96.7	1.147	95.1	1.018	95.3	1.000	95.3	1.001	95.7	0.997	95.7	0.998	95.7	0.994	95.7	0.995
alphafold	50	94.0	1.650	92.7	0.939	88.7	1.000	89.9	1.025	85.5	1.076	86.9	1.103	92.3	1.541	92.8	1.581
alphafold	200	93.7	1.642	94.7	0.988	93.3	1.000	93.3	1.006	93.5	1.034	93.7	1.040	93.1	1.063	93.2	1.069
alphafold	1000	94.3	1.494	95.3	1.015	95.3	1.000	95.3	1.001	95.1	1.005	95.1	1.006	95.0	1.011	95.1	1.012
gene_expression	50	94.6	3.277	94.3	1.000	93.1	1.000	93.8	1.025	94.0	1.115	94.4	1.143	94.1	1.991	94.6	2.042
gene_expression	200	95.1	3.007	95.7	1.023	96.0	1.000	96.2	1.006	95.3	1.033	95.4	1.040	95.7	1.062	95.7	1.068
gene_expression	1000	94.4	2.098	95.3	1.002	95.4	1.000	95.4	1.001	94.7	1.002	94.7	1.003	94.6	1.003	94.6	1.004
census_income	50	86.7	1.297	84.0	1.016	80.9	1.000	81.6	1.025	83.8	1.138	84.6	1.166	85.4	1.179	86.2	1.209
census_income	200	92.4	1.257	91.5	1.019	90.5	1.000	90.5	1.006	91.3	1.065	91.4	1.071	92.0	1.077	92.1	1.083
census_income	1000	94.4	1.207	93.7	1.005	93.5	1.000	93.6	1.001	93.4	1.016	93.4	1.017	93.6	1.026	93.7	1.027
census_healthcare	50	93.9	1.410	94.0	0.997	92.7	1.000	93.4	1.025	92.8	1.065	93.7	1.092	94.2	1.194	94.8	1.225
census_healthcare	200	94.8	1.392	95.3	1.000	95.6	1.000	95.7	1.006	94.6	1.026	94.9	1.032	95.0	1.059	95.0	1.065
census_healthcare	1000	96.5	1.308	95.9	1.017	96.0	1.000	96.2	1.001	95.9	1.006	95.9	1.007	95.1	1.019	95.1	1.020

Table 2: Synthetic: coverage (%) and mean absolute width.

		classical		PPI++		recal-gbm	
setting	n	cov	width	cov	width	cov	width
linear	100	94.8	0.553	95.0	0.393	94.9	0.434
linear	500	94.2	0.248	94.4	0.177	94.9	0.184
linear	2000	95.5	0.124	95.5	0.092	96.1	0.093
nonmono	100	94.2	0.582	93.4	0.568	94.6	0.491
nonmono	500	96.3	0.262	95.8	0.261	94.8	0.141
nonmono	2000	95.3	0.131	95.6	0.131	94.4	0.065
monotone	100	95.2	0.368	94.9	0.234	95.2	0.208
monotone	500	93.9	0.165	94.4	0.106	94.4	0.093
monotone	2000	95.6	0.082	95.5	0.056	95.7	0.049

Findings (real data). *Versus classical:* classical intervals are 1.15–3.3× wider than PPI++ (relative width 1.15–3.28). *Versus recalibration:* recal-iso and recal-gbm were not shorter than PPI++ in any cell beyond Monte-Carlo noise: the only ratios below 1 are forest at $n = 1000$ (0.997, 0.994), and at $n = 50$ the ratios are 1.06–1.14 (iso) and 1.18–2.0 (gbm). At $n = 1000$ they are 1.00–1.03. At $n = 50$ recal-gbm covers better than PPI++ (e.g. alphafold 92.3 vs 88.7) but is wider; recal-iso covers worse on alphafold (85.5). *Coverage:* at $n = 50$ PPI++ under-covers (alphafold 88.7, forest 89.8, census_income 80.9); the classical interval is also poor on census_income (86.7) due to a heavy tail. The t critical value (+2.5% width at $n = 50$) recovers at most 1.2 points. At $n = 1000$ all methods are within about ± 1.5 points of 95% (census_healthcare: PPI++ 96.0, classical 96.5, i.e. slightly conservative). PPI ($\lambda = 1$) was sometimes marginally narrower than PPI++ at $n = 50$ (down to 0.94 on alphafold) but this reflects noisy $\hat{\lambda}$ and was not uniform.

Synthetic check of the mechanism. $Z \sim N(0,1)$, $f = Z$, $N = 20000$, 1000 reps. 'linear': $Y = Z + \varepsilon$ ($\varepsilon \sim N(0,1)$); 'monotone': $Y = \tanh(2Z) + 0.5\varepsilon$; 'nonmono': $Y = Z^2 + 0.5\varepsilon$ (true mean 1; $\rho = 0$, $\eta^2 = 0.89$). Script `code/synth.py`; gbm recalibration. In 'nonmono', PPI++ cannot beat classical ($\lambda^* = 0$; widths 0.131 vs 0.131 at $n = 2000$) while recalibration halves it (0.065) with 94.4% coverage; the oracle ratio from Prop. 1 with $r = 0.1$ is $\sqrt{(1 - 0.89/1.1)/(1 - 0)} \approx 0.44$, versus 0.50 observed. In 'monotone', recalibration gives a smaller gain (0.049 vs 0.056 at $n = 2000$, 12%; 0.208 vs 0.234 at $n = 100$). In 'linear' (already calibrated) it costs 10% at $n = 100$ (0.434 vs 0.393) and 1–4% at larger n. Coverage stays within roughly 93.4–96.3%, i.e. nominal given the ≈ 0.7-point s.e. (PPI++ in 'nonmono' $n = 100$: 93.4).

5 Discussion and limitations

Answer to the question. With f, labels and unlabelled data fixed, recalibrating on the labels can shorten the interval only if the calibration curve of Y given the prediction (and covariates) is substantially nonlinear, and enough labels exist to learn it. The six real predictors here are, as far as widths show, close to linearly calibrated, so PPI++ was already about as short as this route allows, and the learning cost made recal slightly wider. We do not claim no other route could help (e.g. variance-adaptive or bootstrap intervals). **Limitations.** Mean estimation only; no quantiles or regression. The "population" is the empirical distribution of each dataset, which ignores shift (plankton and ballots, which have shift, were not tested). Recalibrator hyperparameters were fixed a priori and not tuned; a better-tuned or more parametric learner could narrow the small-n gap, and we did not test this. The $\hat{\sigma}$ for cross-fitted estimators has no proof; the experiments suggest coverage is acceptable at $n \geq 200$ but at $n = 50$ several methods (including PPI++) under-cover, so those cells should not be read as valid. Widths are reported as averages without MC error bars (coverage has binomial s.e.); the relative-width differences of $\leq 1\%$ should be treated as ties. No multiple-testing correction is involved as no hypothesis tests were run.

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V. Chernozhukov et al. Double/debiased machine learning for treatment and causal parameters. arXiv:1608.00060, 2016.

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A Reproduction (also in README)

Python 3.11, numpy, scipy, scikit-learn, pandas, matplotlib, ppi-python==0.2.3 in a `uv` venv (versions: see `requirements.txt`). Data downloaded by `ppi_py.datasets.load_dataset` to `data/`. Commands: `python run.py 1000` (about 25 min on 40 cores), `python analyze.py` (Table 1), `python synth.py` (Table 2, about 10 min). Seeds: real data $1000r + 7n +$ dataset index; synthetic $r + 1000n$. No literature or pages named as off-limits in the task were accessed.

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